

SAHYADRI PETROCHEMICALS LTD.	Doc No: SPL-CM-VIB-1101A
Reliability & Predictive Maintenance Cell	Rev: - Date: 20-Nov-2024
Classification: INTERNAL	Page 1 of 1

CONDITION MONITORING REPORT

Vibration Survey | Pump-Motor Train P-1101A / M-1101A

Routine vibration survey of pump-motor train **P-1101A / M-1101A**. Overall velocity (RMS) measured at pump drive-end (DE) bearing, evaluated against **ISO 10816-3** for medium machines on rigid foundations.

Survey Date	Overall Vel. (mm/s RMS)	ISO 10816 Zone	Dominant Frequency	Remark
10-Feb-2024	2.8	B (acceptable)	1x RPM	Baseline OK
02-Aug-2024	4.5	C (unsatisfactory)	2x RPM	Rising; 2x indicates misalignment
20-Nov-2024	7.1	D (unacceptable)	2x RPM + harmonics	Severe misalignment signature

Analyst Findings

The dominant **2x running-speed vibration with phase across the coupling** is a classic misalignment signature. The trend has worsened steadily through 2024. The repeated mechanical-seal failures on this pump are consistent with **chronic shaft misalignment** rather than random seal defects.

Recommendation: perform laser alignment of the P-1101A/M-1101A train to OEM tolerance before the next start, and verify the API Plan 52 buffer system is functioning. Replacing the seal alone will not resolve the recurrence.

Reliability Engineer: S. Iyer | Instrument: portable FFT analyser, cal. due 06-2025